

TSA Lab Notes

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Spring 2025

1 Basic Data Preprocessing Techniques in R

1.1 1. Data Loading and Inspection

Before analysis, load the dataset and understand its structure.

```
data = read.csv("your_data.csv")  
head(data)  
str(data)  
summary(data)
```

2. Handling Missing Values

Missing data must be treated appropriately. Common approaches include:

- Removing rows
- Imputing using mean, median, or time-aware methods

```
[language=R]  
data = na.omit(data)  
# Remove rows with missing values  
data$column[is.na(data$column)] = mean(data$column, na.rm = TRUE)  
# Mean imputation
```

3. Converting to Time Series Format

For time series problems, convert data into a **ts** object.

```
[language=R]  
ts_data = ts(data$column, start = c(2000, 1), frequency = 12)  
# Monthly data starting from Jan 2000
```

4. Creating Lagged Variables

Lagged variables help in modeling autocorrelation in time series.

```
data$lag1 = stats::lag(data$column, -1)
```

5. Differencing to Remove Trend

Differencing is used to remove trends or make data stationary.

```
diff_data = diff(ts_data)  # First-order difference
```

6. Creating Survival Objects

For survival analysis, construct survival objects using `Surv()`.

```
library(survival)  
SurvObject = Surv(time = data$duration, event = data$status)
```

7. Censoring Adjustment

Censored data must be coded properly:

- 1 = event occurred
- 0 = right-censored

```
data$status = ifelse(data$event_column == "event", 1, 0)
```

8. Creating Interval Censored Data

For parametric modeling:

```
left = data$duration  
right = data$duration  
right[which(data$status == 0)] = NA  
LRdata = data.frame(left, right)
```

9. Scaling and Normalization (for regression models)

Important for covariates in AFT models.

```
data$scaled_var = scale(data$var)
```

10. Checking Stationarity (for Time Series)

ADF or KPSS tests are used to check stationarity.

```
library(tseries)  
adf.test(ts_data)      # Augmented Dickey-Fuller Test  
kpss.test(ts_data)    # KPSS Test
```

Conclusion

Preprocessing ensures the dataset is clean, correctly formatted, and ready for modeling—whether it's forecasting in time series or estimating hazard functions in survival analysis.